Model Risk Memo: Claim Frequency and Pure Premium Champion/Challenger Review

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1. Purpose and decision requested

This memo documents whether a challenger XGBoost model should replace the champion Poisson GLM for predicting motor third-party liability claim frequency, and whether the result holds once severity is added to form a pure premium. **Recommendation: promote with conditions** (Section 9).

2. Models reviewed

	Champion	Challenger
Frequency model	Poisson GLM, log link, offset = log(Exposure). 53 terms: tree-derived bins for vehicle power, vehicle age, driver age and bonus-malus; a BonusMalus × DrivAge interaction; log density; fuel type; brand; region	XGBoost, count:poisson objective, base margin = log(Exposure). 35 raw features; max_depth 5, eta 0.2, 100 rounds, chosen from a 5-configuration manual grid
Severity model	Gamma GLM (log link) on the same 53 terms	XGBoost, gamma objective (max_depth 3, eta 0.05, 200 rounds, untuned)
Pure premium	Frequency × severity	Frequency × severity

The frequency models were trained by SageMaker Processing and Training jobs (local mode on Docker). The severity models were fit in the validation notebook.

3. Data and validation design

French motor third-party liability data (freMTPL2freq and freMTPL2sev). Frequency: 678,013 policies loaded and 676,789 after cleaning (claim counts capped at 5, exposure capped at 1.0 year). A random 80/20 split (seed 42) gives 541,431 training and 135,358 test policies, about 71,000 exposure-years in the test set. Area was excluded because it overlaps with log density, and rare vehicle brands (B11, B13, B14) were grouped as Other. Severity: claim amounts capped at 100,000 and joined to policies by ID. All models were scored on the identical held-out test set, using the artifacts produced by the training jobs.

4. Performance

Exposure-weighted Gini on the test set, frequency only:

Model	Gini
Champion GLM	0.2976
Challenger XGBoost	0.3348

For context, the GLM improved from 0.2460 (baseline) to 0.2478 (driver-age polynomial terms) to 0.2976 (tree-based binning plus interaction), so the champion is a deliberately strengthened baseline, not a weak one.

- **Lift charts:** both models track actual frequency closely in every decile. The challenger separates risk more sharply: its top decile has an actual frequency of about 0.31 against about 0.27 for the champion, while the lowest deciles are similar (about 0.04). The challenger under-predicts its top decile slightly (about 0.29 predicted vs 0.31 actual).
- **Double-lift chart:** where the models disagree, the challenger is right. In the bucket where XGBoost predicts far lower risk than the GLM, actual frequency is about 0.085, XGBoost predicts about 0.089 and the GLM about 0.165. In the bucket where XGBoost predicts far higher risk, actual is about 0.229, XGBoost about 0.215 and the GLM about 0.125.

Pure premium (frequency × severity) on the same test set:

Model	Pure premium Gini	Predicted ÷ actual total loss (adjusted)
Champion (GLM frequency × GLM severity)	0.259	1.063
Hybrid (XGBoost frequency × GLM severity)	0.302	1.059
Challenger (XGBoost frequency × XGBoost severity)	0.306	1.013

The frequency model drives nearly all of the improvement (+0.043 Gini); XGBoost severity adds only +0.004, which is within noise for a heavy-tailed target. Totals are adjusted for a claim-count mismatch: the test set counts 7,089 claims in the frequency file but only 5,247 have an amount in the severity file (ratio 1.35). The adjustment assumes claims without an amount resemble those with one. Gini is unaffected because a constant scale does not change rankings.

5. Explainability

SHAP values were computed for the challenger on the test set (log-frequency scale). Mean absolute contribution ranks BonusMalus first (about 0.29), then vehicle age (0.18), driver age (0.14), log density (0.07), vehicle power (0.05) and brand B12 (0.05). Region indicators are individually small (each under about 0.02), so geography enters mainly through density. This ranking agrees with the GLM coefficients and with a separate decision-tree analysis. XGBoost's gain-based importance overstated brand B12 and fuel type, so the SHAP ranking is the one to rely on.

Per-policy reason codes can be produced from the same values. For the highest-risk test policy, the top risk-raising factors were a brand-new vehicle (about 5.5× on claim frequency), vehicle power (1.5×) and BonusMalus 75 (1.4×). Because SHAP spreads interaction effects across features, these multipliers do not multiply exactly to the policy's total risk ratio.

6. Fairness and proxy review

The data contains no protected-class attributes, so this is a proxy review on population density and region, using test-set predictions. Regions with under 1,000 exposure-years in the test set (8 regions) were excluded from the region table.

Calibration by density band (predicted ÷ actual frequency, 1.0 is ideal):

Density band	Actual frequency	GLM	XGBoost
Q1 rural	0.082	1.035	1.040
Q2	0.092	1.025	1.018
Q3	0.105	0.997	1.005
Q4 urban	0.124	1.010	1.003

Disparate impact ratio (80% rule), computed on the share of policies not placed in the top-20% risk tier:

Grouping	GLM	XGBoost
Density band	0.821	0.813
Region	0.777	0.789

Interpretation. Both models clear 0.80 on density. The region ratio is just under 0.80 for both models and is set by two groups: the reference region (surcharge share about 31-32%, actual frequency 0.132, the highest of any region) and R41 (surcharge share about 13%, actual frequency 0.081, the lowest). The reference region is priced accurately (calibration 1.006 GLM, 1.010 XGBoost), so the gap reflects observed claims, not over-penalization. Calibration falls outside ±10% for R26 (about 1.18 for both models, under 100 claims in the test set), R52 (about 1.12 for both) and, for XGBoost only, R31 (1.13 vs 1.06 for the GLM). Elsewhere the two models deviate in the same direction by similar amounts, so the challenger adds little new miscalibration. R31 is flagged for monitoring.

7. Monitoring and governance controls

- **Drift monitoring.** A PSI and KS/chi-squared monitor treats the training data as the frozen baseline and flags PSI above 0.2. On the real test set every feature is stable (PSI near 0, score PSI 0.0002), as expected for a random split. On a deliberately shifted copy it flags vehicle age (PSI 0.30), driver age (0.34) and BonusMalus (6.2, inflated because most policies sit at the floor value of 50) and the model score (3.8), and issues a retrain-and-revalidate decision. This covers covariate drift only; detecting concept drift needs fresh labelled outcomes.
- **Model Registry.** Both models are registered as versions of one SageMaker model package group: version 1 (champion, Approved) and version 2 (challenger, Approved with conditions). Each version records its metrics, the data fingerprints (S3 ETags) for the train and test files, and the training job name.
- **Automated promotion gate.** GitHub Actions runs unit tests and a gate on every push. The gate requires a Gini improvement of at least 0.01, a pure premium Gini that is no worse, disparate impact ratios within 0.03 of the champion's and at least 0.70, and pure premium calibration within ±10%. The challenger passes all checks. The fairness rule is relative to the champion because the region gap reflects real claim-rate differences, and an absolute 80% threshold would fail both models.

8. Limitations

- Challenger frequency hyperparameters were selected using test-set Gini, with no separate validation set or cross-validation. The depth-5 configurations were within 0.0004 Gini of each other, so the optimism is likely small, but the test Gini is not a fully clean estimate.
- One random split; no confidence intervals on Gini or calibration.
- The severity models are untuned and add little; the pure premium totals depend on the claim-count adjustment described in Section 4.
- Proxy-only fairness review. The 80% rule is a screening heuristic, and a ratio below 0.80 is not by itself a finding of discrimination.
- The pipeline ran in SageMaker local mode (Docker) because the account's cloud compute quota was 0. There is no deployed endpoint, no SageMaker Model Monitor and no SageMaker Clarify; the local drift script and the notebook's fairness and SHAP analysis stand in for them.

9. Recommendation and conditions

Promote the XGBoost challenger to production candidate, subject to these conditions:

1. Re-evaluate on a fresh holdout or with cross-validation, with confidence intervals, before use.
2. Monitor regional calibration, especially R52, R26, R31 and R93.
3. Investigate why about a quarter of counted claims have no severity record, and tune the severity model, before relying on pure premium.
4. Run the drift monitor on live data, and add a concept-drift check against labelled outcomes, before deployment.
5. Have actuarial and compliance reviewers approve the SHAP-based reason codes before any adverse-action use.

10. Sign-off (simulated)

Role	Name	Decision	Date
Model developer		Submitted	
Actuarial reviewer			
Compliance reviewer			